

$$= \int_0^2 2 - \frac{8}{u^2+4} du$$

$$= \int_0^2 2 du - 4 \int_0^2 \frac{2}{u^2+2^2} du.$$

$$= [2u]_0^2 - 4 \left[\tan^{-1}\left(\frac{u}{2}\right) \right]_0^2.$$

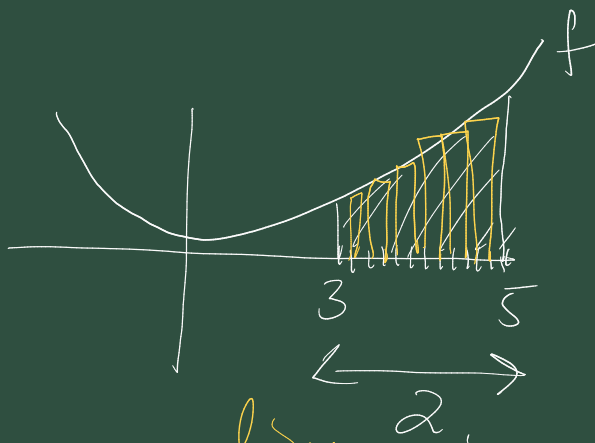
$$= 4 - 4 \left[\underbrace{\tan^{-1}(1)}_{\pi/4} - \underbrace{\tan^{-1}(0)}_0 \right]$$

$$= 4 - 4 \frac{\pi}{4}$$

$$= 4 - \pi$$



(b)



$$\frac{98}{3}$$

lim
width $\rightarrow 0$
no. of rectangles $\rightarrow \infty$

using the equally spaced partition
 $x_i = 3 + i \frac{2}{n}, \quad i = 0, \dots, n$

